Sahaj Bhadja

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Summary

Software development engineer with 2+ years building **Java/Python software components**, SaaS applications, and financial data systems; strong in data structures, algorithms, object-oriented design, debugging, code review, documentation, teamwork, and continuous improvement.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and implemented **reusable Python software components** and normalized relational schemas for 500+ users, applying object-oriented design, indexed queries, and optimized access patterns to reduce latency from 1.2 seconds to 350 milliseconds.
- Debugged and maintained **production data workflows** across 5,000+ monthly jobs, adding validation, reconciliation, error handling, documentation, and CI/CD releases to achieve 99%+ reliability while learning evolving business requirements.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **full-stack React and Flask application components** for uploads, processing, history, and reporting, supporting 150+ users and 10,000+ daily requests through modular MVC architecture and reliable REST services.
- Collaborated with peers across **design, coding, testing, deployment, and support**, incorporating review feedback, debugging file-processing defects, and delivering GitHub Actions-backed enhancements on schedule while reducing feedback turnaround time 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Implemented **object-oriented software and API components** using Singleton and Factory patterns, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Actively participated in **code reviews and Agile collaboration** with Microsoft and GitHub stakeholders, automating builds, deployments, and 20+ tests through Jenkins, PyTest, and Mockito across three operating systems.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed and managed **React, Node.js, and Azure SQL components** across eight SaaS services for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs and WebSockets.
- Improved **software quality and operational reliability** through Jest, Selenium, Sentry, Azure Monitor, Docker packaging, and Jenkins CI/CD, cutting deployment time 50% while constructively resolving defects with teammates.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Spring Boot and Flask financial components** for bookings, pricing, payments, and status updates, integrating Firebase persistence and validated webhooks while maintaining sub-200-millisecond responses across 50,000+ application records.
- Analyzed client needs and **documented end-to-end software workflows**, mapping 12+ integrations and delivering unit-tested, version-controlled enhancements that improved cross-platform compatibility 15%, delivery speed 22%, and client satisfaction.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
Computer Science: Data structures, algorithms, OOP, design patterns, multithreading, problem solving
Software: Spring Boot, FastAPI, Flask, Node.js, React.js, REST APIs, microservices, SaaS
Data & Cloud: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, AWS, Azure, Docker, Kubernetes
Engineering: Debugging, testing, code review, documentation, Git, Jenkins, GitHub Actions, CI/CD, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **financial SaaS platform** supporting 50,000+ users and 1M+ transactions with secure bank integrations, real-time synchronization, advanced filtering, MongoDB data access, responsive interfaces, and Kubernetes-based CI/CD automation.

Chat2SQL Code | *Java, TypeScript, Oracle, Docker, Maven*
- Enhanced an **AI-powered database application** across Java and TypeScript components, supporting Oracle and 10+ SQL/NoSQL systems with Maven, Docker, schema exploration, query generation, and cross-platform debugging workflows.

CPU Scheduling Simulator Live | *C++, React.js, Data Structures, Algorithms*
- Implemented and tested **six scheduling algorithms** with React controls, Gantt visualization, edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.